

What we're building today

A quick brief before we open the editor: what it does, what we're using to build it, and how it's wired together.

01 Brief

Rafiq v1 is a simple chatbot. You ask it something, it replies — that's it. No memory of past conversations, no company data, no extra logic. We're keeping it this basic on purpose, so we can clearly see later exactly where it breaks and why.

02 Packages

Five small tools, each doing one job:

FastAPI

Builds the backend — the part that receives a question and sends back an answer.

Uvicorn

Actually runs the FastAPI app as a live server.

OpenAI SDK

Lets us talk to GPT without handling the raw API calls ourselves.

python-dotenv

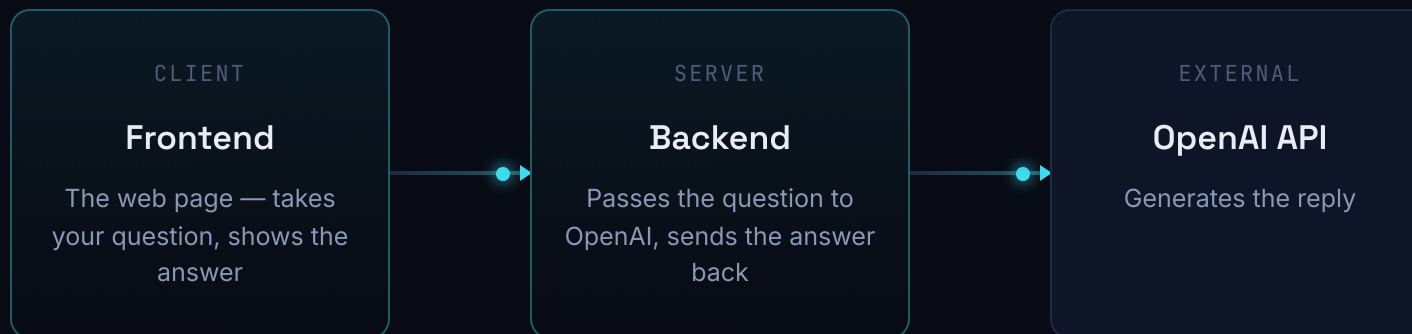
Keeps the API key in its own file, out of the code.

Pydantic

Checks that the incoming question is valid before we send it anywhere.

03 Architecture

Three pieces. A page you type into, a small server in the middle, and OpenAI doing the actual thinking.



one question in, one answer back — nothing else happening yet

04 Tokens & cost

A **token** is just a small chunk of text — sometimes a whole word, sometimes part of one. "Hello" is one token, but a longer or less common word might get split into two or three.

Every question you send and every answer you get back is counted in tokens, and that's exactly what you pay for — input tokens and output tokens, each at its own rate. More tokens in the conversation simply means a higher cost for that request.